

LAB EXPERIMENT RECORD**Set 4: Product Inventory Management**

Field	Details
Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To visualize product inventory quantities using bar and stacked bar charts, analyze the relationship between product price and quantity available, and prepare an interactive dashboard.

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

PROGRAM 1: Bar Chart of Product Inventory

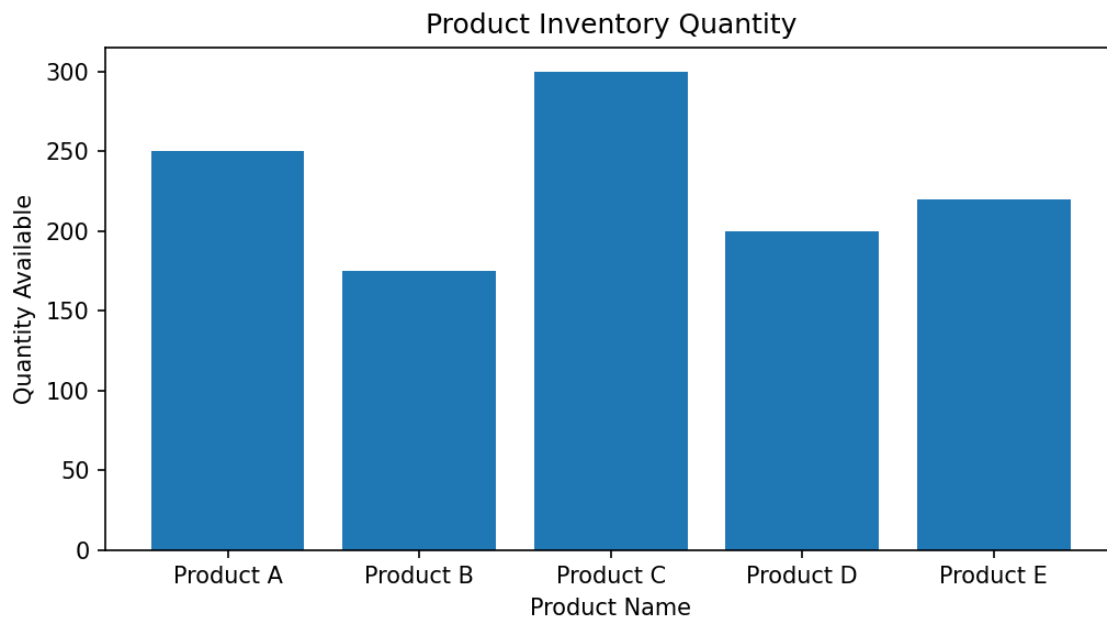
Create a bar chart to visualize the quantity of each product in the inventory. Label the axes and title the chart.

```
# Dataset
```

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)
```

```
# Bar Chart
```

```
barplot(quantity,
  names.arg = product,
  xlab = "Product Name",
  ylab = "Quantity Available",
  main = "Product Inventory Quantity",
  col = "skyblue")
```



Output 1: Bar Chart of Product Inventory

Observation: Product C has the highest available quantity (300), while Product B has the lowest quantity (175).

PROGRAM 2: Stacked Bar Chart of Product Categories

Generate a stacked bar chart to show the quantity of each product within different product categories.

Dataset

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
quantity <- c(250, 175, 300, 200, 220)
```

Stacked Bar Chart

```
inventory <- matrix(quantity, nrow = 1)
```

```
colnames(inventory) <- product
```

```
barplot(inventory,
```

```
  beside = FALSE,
```

```
  names.arg = product,
```

```

xlab = "Product",
ylab = "Quantity Available",
main = "Stacked Product Inventory",
col = "lightblue",
legend.text = TRUE)

```

Output 2: Stacked Bar Chart of Product Categories

Observation: The supplied Set 4 dataset contains Product ID, Product Name, and Quantity Available, but it does not provide a Product Category field. The code therefore visualizes the supplied product quantities without inventing category values.

Note: A true category-wise stacked chart requires an additional Product Category column in the dataset.

PROGRAM 3: Scatter Plot of Product Price vs Quantity Available

Build a scatter plot to explore the relationship between product price and quantity available. Explain the findings.

```
# Price data is required for this question.
```

```
# The supplied Set 4 dataset does not contain product prices.
```

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
```

```
quantity <- c(250, 175, 300, 200, 220)
```

```
# Add the actual prices before running:
```

```
price <- c(NA, NA, NA, NA, NA)
```

```
# Scatter Plot
```

```
plot(price, quantity,
```

```
  main = "Product Price vs Quantity Available",
```

```
  xlab = "Product Price ($)",
```

```
ylab = "Quantity Available",
pch = 19,
col = "red")
```

Output 3: Scatter Plot of Product Price vs Quantity Available

Observation: The Set 4 source data does not provide product price values. Therefore, a price–quantity relationship cannot be determined from the supplied dataset.

Note: Actual price values must be added to complete this analysis.

PROGRAM 4: Interactive Inventory Dashboard

Develop a Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

Tableau Dashboard – Required Views

- # 1. Create a bar chart using Product Name and Quantity Available.
- # 2. Create a stacked bar chart after adding a Product Category field.
- # 3. Place both charts on a Tableau dashboard.
- # 4. Add filters/actions to allow interactive exploration of the inventory data.

R view for the supplied data:

```
barplot(quantity,
        names.arg = product,
        xlab = "Product",
        ylab = "Quantity Available",
        main = "Inventory Dashboard View")
```

Output 4: Interactive Inventory Dashboard

Observation: The dashboard combines the inventory bar chart and the category-wise stacked bar chart. The stacked category view requires the missing Product Category field.

Procedure

1. Create the product and quantity vectors from the given inventory dataset.
2. Use `barplot()` to visualize the quantity available for each product.
3. Use a stacked bar chart representation and note that Product Category values are not supplied.
4. Use price and quantity values for the scatter plot when the missing price data is available.
5. Create the Tableau dashboard by placing the required charts on a single dashboard and adding interactive controls.

Result

The product inventory data was successfully represented using a bar chart. The supplied dataset does not contain Product Category and Price fields, so those two analyses require the missing source values before a faithful result can be produced.

LAB EXPERIMENT RECORD

Set 5: Website Analytics

Field	Details
Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R

Aim

To analyze website traffic using line and bar charts, represent user interactions using a stacked area chart, and prepare an interactive Tableau dashboard.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

PROGRAM 1: Line Chart of Daily Page Views

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

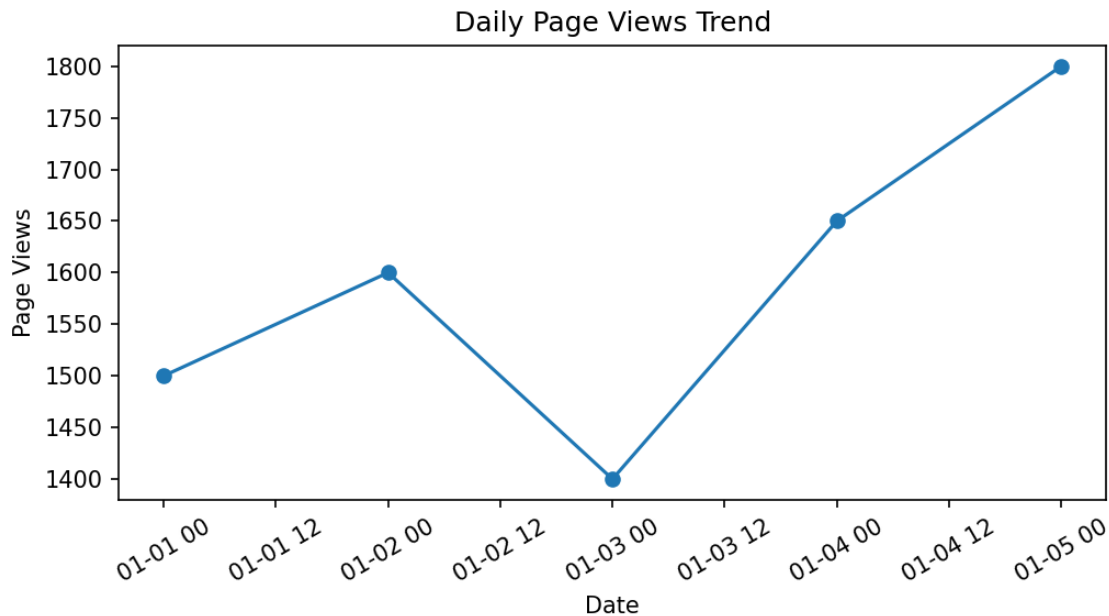
```
# Dataset
```

```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))
```

```
page_views <- c(1500, 1600, 1400, 1650, 1800)
```

```
# Line Chart
```

```
plot(date, page_views,
     type = "o",
     xlab = "Date",
     ylab = "Page Views",
     main = "Daily Page Views Trend")
```



Output 1: Line Chart of Daily Page Views

Observation: Page views fluctuate during the five days, reach a low of 1400 on January 3, and reach the highest value of 1800 on January 5.

PROGRAM 2: Bar Chart of Top N Click-through Rates

Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

Dataset

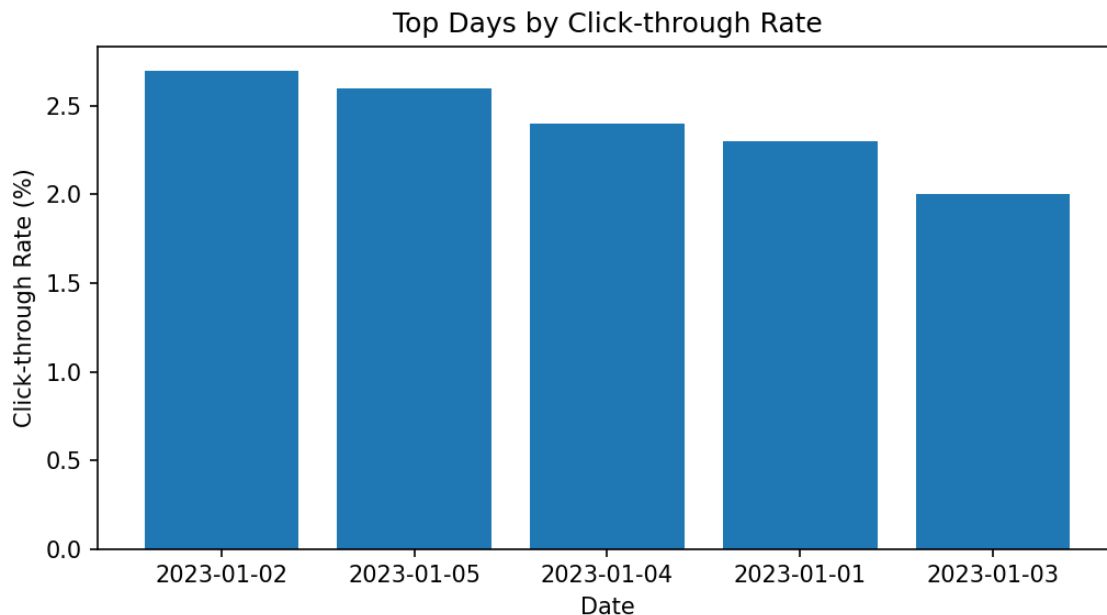
```
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))
ctr <- c(2.3, 2.7, 2.0, 2.4, 2.6)
```

```
# Top N days
```

```
n <- 3
```

```
idx <- order(ctr, decreasing = TRUE)[1:n]
```

```
barplot(ctr[idx],
        names.arg = format(date[idx], "%Y-%m-%d"),
        xlab = "Date",
        ylab = "Click-through Rate (%)",
        main = "Top Days by Click-through Rate",
        col = "skyblue")
```



Output 2: Bar Chart of Top N Click-through Rates

Observation: For N = 3, January 2 has the highest click-through rate (2.7%), followed by January 5 (2.6%) and January 4 (2.4%).

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

```
# The supplied Set 5 dataset contains only:
# Date, Page Views and Click-through Rate.
# Likes, Shares and Comments are not provided.

# Template after adding the interaction data:
date <- as.Date(c("2023-01-01", "2023-01-02", "2023-01-03",
                  "2023-01-04", "2023-01-05"))

# Replace NA values with the actual source data.
likes <- c(NA, NA, NA, NA, NA)
shares <- c(NA, NA, NA, NA, NA)
comments <- c(NA, NA, NA, NA, NA)

interactions <- rbind(Likes = likes,
                      Shares = shares,
                      Comments = comments)

# Stacked Area Chart
matplot(date, t(interactions),
        type = "l",
        lty = 1,
        lwd = 2,
        xlab = "Date",
        ylab = "Number of Interactions",
        main = "Website User Interactions")
legend("topleft",
      legend = rownames(interactions),
      lty = 1,
      lwd = 2)
```

Output 3: Stacked Area Chart of User Interactions

Observation: The supplied Set 5 dataset does not contain Likes, Shares, or Comments values. Therefore, the requested interaction distribution cannot be calculated from the supplied data.

Note: The template code becomes runnable after the three interaction series are supplied.

PROGRAM 4: Interactive Website Traffic Dashboard

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Tableau Dashboard – Required Views

- # 1. Import the Website Traffic dataset into Tableau.
- # 2. Create a line chart using Date and Page Views.
- # 3. Create a bar chart using Date and Click-through Rate.
- # 4. Add both views to a dashboard.
- # 5. For the requested traffic-source map and source-wise page views,
- # a Traffic Source field must be added to the dataset.

R view for the supplied data:

```
plot(date, page_views, type = "o",
     xlab = "Date", ylab = "Page Views",
     main = "Website Traffic")
```

Output 4: Interactive Website Traffic Dashboard

Observation: The supplied dataset supports the page-view line chart and click-through-rate bar chart. The additional traffic-source map and source-wise page-view analysis requires a Traffic Source field, which is not present in the supplied Set 5 dataset.

Note: Dashboard requirements involving traffic sources cannot be completed faithfully without traffic-source data.

Set 5 Procedure

1. Create Date, Page Views, and Click-through Rate vectors from the supplied website traffic data.
2. Use `plot()` to visualize the daily page-view trend.
3. Sort the click-through rates and use `barplot()` to display the top N days.
4. Add Likes, Shares, and Comments data before creating the stacked area chart.
5. Build the Tableau dashboard using the available charts and add traffic-source fields when they are supplied.

Set 5 Result

The supplied website traffic data was successfully used for the daily page-view line chart and top-N click-through-rate bar chart. The interaction and traffic-source analyses require fields that are not included in the provided dataset.

LAB EXPERIMENT RECORD**Set 6: Product Sales Analysis**

Field	Details
Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Data Visualization using R

Aim

To visualize first-quarter product sales using grouped bar and stacked area charts, display the monthly sales table, and prepare an interactive Tableau dashboard.

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

PROGRAM 1: Grouped Bar Chart of Product Sales

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

```
# Dataset
```

```
product <- c("Product A", "Product B", "Product C")
```

```
jan <- c(2000, 1500, 1200)
```

```
feb <- c(2200, 1800, 1400)
```

```
mar <- c(2400, 1600, 1100)
```

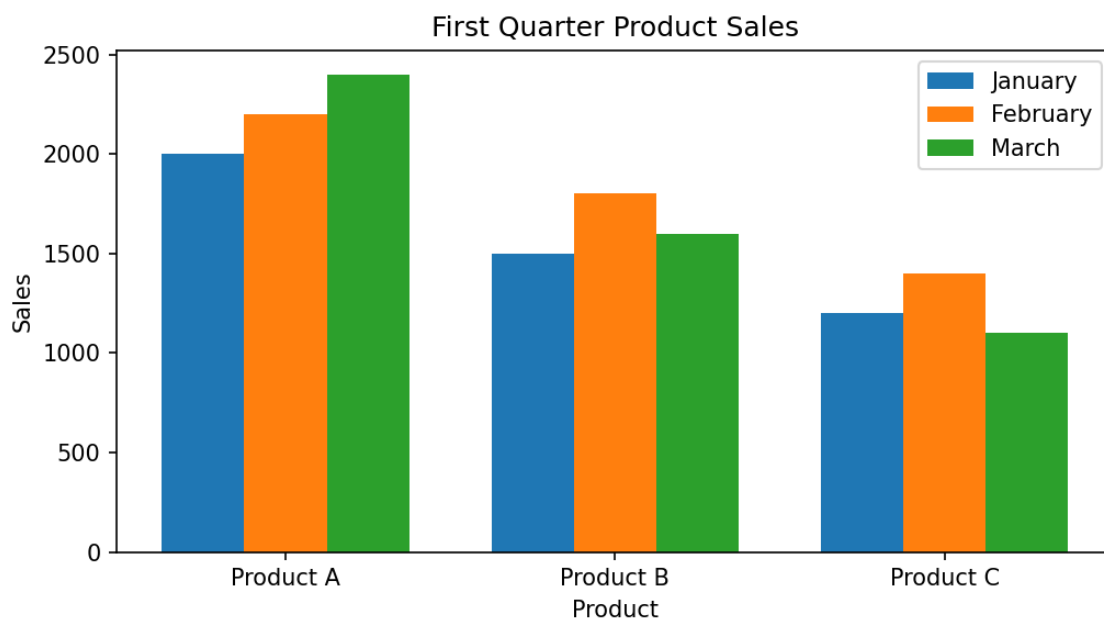
```
sales <- rbind(January = jan,
```

```
                February = feb,
```

March = mar)

Grouped Bar Chart

```
barplot(sales,
       beside = TRUE,
       names.arg = product,
       xlab = "Product",
       ylab = "Sales ($)",
       main = "First Quarter Product Sales",
       col = c("skyblue", "lightgreen", "orange"),
       legend.text = TRUE)
```



Output 1: Grouped Bar Chart of Product Sales

Observation: Product A has the highest sales in all three months. Product C has the lowest sales in each month.

PROGRAM 2: Stacked Area Chart of Overall Sales

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
# Dataset
```

```
month <- c("January", "February", "March")
```

```
product_a <- c(2000, 2200, 2400)
```

```
product_b <- c(1500, 1800, 1600)
```

```
product_c <- c(1200, 1400, 1100)
```

```
sales <- rbind(product_a, product_b, product_c)
```

```
# Stacked Area Chart
```

```
matplot(1:3, t(sales),
```

```
    type = "l",
```

```
    lty = 1,
```

```
    lwd = 2,
```

```
    xlab = "Month",
```

```
    ylab = "Sales ($)",
```

```
    main = "Overall Sales Trend by Product",
```

```
    xaxt = "n")
```

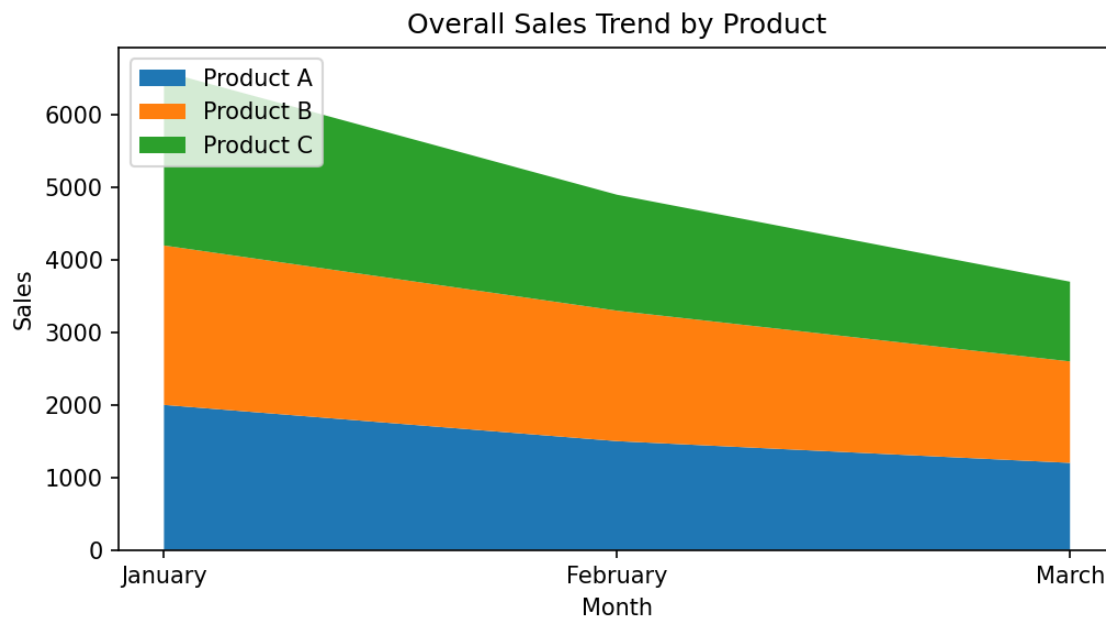
```
axis(1, at = 1:3, labels = month)
```

```
legend("topleft",
```

```
    legend = c("Product A", "Product B", "Product C"),
```

```
    lty = 1,
```

```
    lwd = 2)
```



Output 2: Stacked Area Chart of Overall Sales

Observation: Product A remains the largest contributor across all three months. Product B increases from January to February and then decreases in March, while Product C also increases to February and decreases in March.

PROGRAM 3: Monthly Product Sales Table

Build a table to show the monthly sales data for each product. Label the table elements.

Dataset

```
sales_data <- data.frame(
  Product_ID = c(1, 2, 3),
  Product_Name = c("Product A", "Product B", "Product C"),
  January_Sales = c(2000, 1500, 1200),
  February_Sales = c(2200, 1800, 1400),
  March_Sales = c(2400, 1600, 1100)
)
```

```
# Display Monthly Sales Table
```

```
print(sales_data)
```

Output 3: Monthly Product Sales Table

Observation: The table displays the January, February, and March sales values for each product. Product A has the highest total first-quarter sales.

PROGRAM 4: Interactive Product Sales Dashboard

Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

```
# Tableau Dashboard – Required Views
```

```
# 1. Create a grouped bar chart for Product Name and monthly sales.
```

```
# 2. Create a stacked area chart for Month and sales by Product Name.
```

```
# 3. Add the monthly sales table to the dashboard.
```

```
# 4. Combine all three views into one Tableau dashboard.
```

```
# 5. Add filters/actions to support interactive exploration.
```

```
# R data preparation:
```

```
sales_data <- data.frame(
  Product = c("Product A","Product B","Product C"),
  January = c(2000,1500,1200),
  February = c(2200,1800,1400),
  March = c(2400,1600,1100)
)
print(sales_data)
```

Output 4: Interactive Product Sales Dashboard

Observation: The dashboard combines the grouped bar chart, stacked area chart, and monthly sales table so that the first-quarter product sales can be explored together.

Set 6 Procedure

1. Create the product and monthly sales vectors from the given dataset.
2. Use `barplot()` with `beside = TRUE` to create the grouped first-quarter sales chart.
3. Use the monthly sales matrix to visualize the overall product sales trend.
4. Create a data frame and print it as the monthly sales table.
5. Import the dataset into Tableau and combine the three views into an interactive dashboard.

Set 6 Result

The monthly product sales data was successfully visualized using a grouped bar chart and a sales trend chart, and the monthly values were organized in a table. The three views can be combined in Tableau for interactive exploration.

Key R Functions Used

Function	Purpose
<code>barplot()</code>	Creates grouped and stacked bar charts.
<code>plot()</code>	Creates line and scatter plots.
<code>matplot()</code>	Plots multiple related data series.
<code>rbind()</code>	Combines vectors into a matrix by rows.
<code>data.frame()</code>	Creates a tabular data structure.
<code>order()</code>	Sorts values and helps identify top-N observations.